

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Non-interventional Study for Kisqali (Ribociclib) in Combination With an Aromatase Inhibitor for Adjuvant Treatment in Patients With HR+/HER2- Early Breast Cancer at High Risk of Recurrence **Short Title/Acronym:** CAROLEEN **Protocol Number:** CLEE01101DE01 **NCT Number:** NCT06830720 **Posting ID:** 733112565215622474 **Sponsor/Company Name:** Novartis **Investigational Product:** Kisqali (ribociclib) + Aromatase Inhibitor (AI) ± LHRH **Phase of Development:** Not Applicable (Observational / Non-interventional Study) **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world effectiveness and safety profile of ribociclib in combination with an aromatase inhibitor for adjuvant treatment in patients with HR+/HER2- early breast cancer at high risk of recurrence.

Secondary Objectives: To evaluate patient compliance, quality of life (QoL), management of adverse events, and analyze the clinical adoption of ribociclib + AI ± LHRH over time by comparing cohorts with abemaciclib + ET ± LHRH and ET monotherapy ± LHRH.

2.2 Background & Rationale To provide information on the real-world effectiveness, safety, and tolerability of ribociclib combined with an AI in the adjuvant setting for HR+/HER2- early breast cancer. Gathering insights into the potential benefits and risks associated with these regimens in routine practice will inform clinical decision-making and contribute to improved patient outcomes across the healthcare ecosystem.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational / Non-interventional Study **Control Type:** Active Comparator cohorts (abemaciclib (Verzenio) + ET ± LHRH, and ET mono ± LHRH)

Blinding: Open-label (None) **Randomization:** N/A (Treatment decision based solely on clinical judgement prior to enrollment)

3.2 Planned Enrollment & Duration Targeted Enrollment: 3250

Number of Sites: 267 locations (German and Austrian breast centers and gynecological practices) **Estimated Study Start:** 20-Feb-2025 **Estimated Primary Completion:** 30-Jun-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histological diagnosis of HR+/HER2-early breast cancer with curative intent. 2. Patients must have an indication for treatment with ribociclib + AI $\pm$ LHRH, or abemaciclib + ET $\pm$ LHRH, or ET monotherapy as described in the current Summary of Product Characteristics (SmPC) in the adjuvant setting. 3. Before enrollment, the treating physician has made the decision in accordance with the patient to treat them with one of the study regimens, and baseline is no longer than 2 weeks (14 days) prior to written informed consent. (Baseline = date of therapy start for ribociclib or abemaciclib cohorts; within 4 weeks after therapy start or within 4 weeks after last non-endocrine based therapy for the ET mono cohort). 4. $\geq$ 18 years of age. 5. Written informed consent.

4.2 Exclusion Criteria 1. Patient is simultaneously participating in any investigational trial or simultaneously participating in another Novartis-sponsored non-interventional study with ribociclib. 2. Contraindications as per the local SmPC of the prescribed therapy. 3. Patients without a finalized treatment decision prior to enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ribociclib + AI $\pm$ LHRH or comparator therapies (abemaciclib [Verzenio], ET monotherapy) dosed according to routine clinical practice and current effective local SmPC. **Route of Administration:** Oral (Ribociclib/abemaciclib/AI) and Injection (LHRH, if applicable). **Duration of Treatment:** Long-term adjuvant treatment as determined by the treating physician in routine practice.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for co-morbidities as prescribed by the healthcare provider. **Prohibited:** Any therapies strictly conflicting with the SmPC of the prescribed regimen.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Baseline	Day 0	Informed Consent, Medical History, Treatment Decision Documentation, Socio-economic Data Collection
Observation	Routine Care	QoL Questionnaires, Compliance Tracking, Safety & Effectiveness Monitoring
End of Study	Study Completion	Final Data Collection, Exit Documentation

(Note: Visit schedules are dictated by routine clinical practice rather than a mandated trial protocol).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Invasive disease-free survival (iDFS) for adjuvant therapy with ribociclib + AI $\pm$ LHRH in participants with HR+/HER2- eBC at high risk of recurrence using STEEP (Standardized Definitions for Efficacy Endpoints in Adjuvant Breast Cancer Trials) criteria, as assessed by the investigator. **Measurement Method:** iDFS is defined as the time from study start to the date of the first event of invasive ipsilateral breast tumor recurrence, local/regional invasive recurrence, distant recurrence, death (any cause), contralateral invasive BC, or second primary non-breast invasive cancer (excluding basal and squamous cell carcinomas of the skin).

7.2 Secondary & Exploratory Endpoints * Number of participants per Baseline parameters. * Reasons for treatment decision by treating physician. * Patients' individual perception of risk of recurrence and treatment decision. * Patient Quality of Life (QoL). * Patient compliance to therapy. * Management of adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via HCP documentation during routine clinical visits and observational data capture. **Serious Adverse Events (SAEs):** Reported according to local pharmacovigilance regulations and the Novartis drug safety guidelines. **Data Monitoring Committee (DMC):** Internal non-interventional registry monitoring board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Cross-sectional and longitudinal analyses of distinct cohorts (ribociclib + AI $\pm$ LHRH; abemaciclib + ET $\pm$ LHRH; ET monotherapy $\pm$ LHRH). **Sample Size Justification:** Designed with a targeted enrollment of 3250 participants to describe clinical routine in a representative subset of the German/Austrian healthcare eco-system. **Handling of Missing Data:** Standard imputation methodology appropriate for real-world/observational datasets.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local regulations for non-interventional studies. **Monitoring Plan:** Centralized and selective onsite/remote data verification to ensure registry data

validity. **Audit Trail:** eCRF locking and compliance with real-world data security standards.

11. Study Identifiers & Governance

Primary ID: CLEE01101DE01 **Registry Number:** NCT06830720 **Posting ID:** 733112565215622474 **Sponsor/Lead Organization:** Novartis **Study Title:** CAROLEEN **Official Regulatory Title:** A Non-interventional Study for Kisqali (Ribociclib) in Combination With an Aromatase Inhibitor for Adjuvant Treatment in Patients With HR+/HER2- Early Breast Cancer at High Risk of Recurrence to Evaluate Real-world Effectiveness, Safety Profile, Patient Compliance and Quality of Life

12. Clinical Framework (Breast Cancer Specific)

Disease Area / Primary Condition: Breast Neoplasms (HR+/HER2- Early Breast Cancer) **Preferred UMLS Name:** Adjuvant Therapy for Hormone Receptor Positive HER2 Negative Early Breast Cancer **Diagnosis Threshold:** Histological diagnosis of HR+/HER2- early breast cancer at high risk of recurrence **Medical Methodology:** Adjuvant Endocrine Therapy + CDK4/6 Inhibitor **Optimization Target:** Maintenance of Quality of Life and Recurrence risk reduction (iDFS)

13. Study Procedures & Methodology

Management Pathway: Routine clinical pathways for adjuvant breast cancer treatment. **Education Protocol (HCP):** Local SmPC adherence **Education Protocol (Patient):** Standard clinical counseling **Quality Audit Mechanism:** Assessment of patient compliance and real-world data fidelity

14. Observation & Follow-Up Schedule

Total Duration: Estimated 5+ years (Follow-up until estimated completion in 2030) **Onsite Visit Cadence:** As per routine clinical practice **Remote Monitoring Frequency:** Routine data entry schedules corresponding to site visits **Checklist Review Interval:** Periodic Quality of Life (QoL) and socio-economic data collection surveys

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |